

# LESSON 4.3

## APPLYING EXPONENT LAWS



### LESSON INTRODUCTION

**MA.8.NSO.1.3** Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

### LEARNING TARGETS

- I can apply the laws of exponents to generate equivalent numerical expressions.

### BENCHMARK COHERENCE

#### BUILDING ON

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

#### WORKING TOWARD

**MA.912.NSO.1.1** Extend previous understanding of laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

### ESSENTIAL QUESTIONS

- How do you apply multiple exponent laws to generate equivalent numerical expressions?

### LESSON NARRATIVE

Students collaboratively practice using multiple exponent laws to generate equivalent numerical expressions involving rational bases and integer exponents. Through the activities, students gain experience with using different laws and/or a different sequence in applying exponent laws to generate an expression that is equivalent. Students individually demonstrate their conceptual understanding and procedural fluency after several activities that involve reflective practice.

### COMPONENT SUMMARY

#### 4.3.1 WARM-UP (~5 MIN.)

Write equivalent expressions using only digits zero through nine

#### 4.3.3 ACTIVITY (15 MIN.)

Take turns in teams applying exponent laws to numeric expressions

#### 4.3.5 YOUR TURN! (5-10 MIN.)

Rewrite numerical expressions to have one base and a positive exponent using the laws of exponents

#### 4.3.2 COLLABORATION (10 MIN.)

Analyze and complete student work for using exponent laws to write equivalent expressions

#### 4.3.4 COLLABORATION (10-15 MIN.)

Identify if two expressions are equivalent and use the laws of exponents to justify

#### 4.3.6 WRAP-UP (~5 MIN.)

Reflect on current understanding of the learning target

## RESOURCES

### GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

### MATERIALS

- (Optional) Highlighters
- (Optional) Student whiteboards

### ADDITIONAL PREPARATION

- Use observational data from prior lessons along with 4.2.6 Ticket Out the Door from Lesson 2 to plan for strategic grouping to use during the lesson.

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may struggle to understand how to apply the laws of exponents to a decimal raised to a negative exponent. For example:  $(2.5)^{-2} = (\frac{2}{5})^2$ .
- Students may believe that the laws of exponents must be applied in a specific order. 4.3.2 Collaboration reinforces for students that laws can be applied in a different order and result in the same value.

### THINGS TO CONSIDER

- Review how to convert decimals to their equivalent fractions.
- Expressions can be simplified by applying the laws of exponents in different orders. During debrief discussions, ask students who applied the laws in different sequences to share their work.
- If student whiteboards are accessible, consider allowing teams to use them for the 4.3.3 Activity.

### LITERACY AND LANGUAGE ROUTINES

#### Group Presentations

Students work with partners during the 4.3.2 Collaboration, 4.3.3 Activity, and 4.3.4 Collaboration. Select groups or partner pairs to share one of the questions from each component. This provides an opportunity for students to verbally articulate their group thinking and justification in the laws they chose to use. Engage all students by asking if they used the same order (or same steps). If a group used different steps or a different order, have someone from that group share. This will help reinforce for students there is more than one way to apply the laws of exponents to generate an equivalent numerical expression.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.3.1 Mathematical Fluency

4.3.3 Activity and 4.3.4 Collaboration provide students multiple opportunities to practice writing equivalent expressions. Both components give students flexibility to apply exponent laws of their choosing and have students explain their choices. This repeated and varied practice helps students develop conceptual understanding of and procedural fluency with the learning target. Students have the chance to show their mathematical fluency independently in the 4.3.5 Your Turn!

## DIFFERENTIATE INSTRUCTION

4.3.3 Activity provides auditory learners and visual learners the opportunity analyze team members' work (visually) and discuss (verbally) before completing the next step. Students get to explain verbally and synthesize by writing the steps taken and exponent laws applied. Within the activity, students get a chance to enter each question at a different step, giving all students practice with different entry points.

Additional differentiation opportunities are denoted throughout the lesson guide.

| Team Member 1: _____                                                                                                                                                                                                                  |                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| $\left(\left(\frac{4}{5}\right)^0 \cdot \left(\frac{5}{4}\right)^{-2}\right)^3$ $\left(\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^2\right)^3$ $\left(\left(\frac{4}{5}\right)^2\right)^3$ $\left(\frac{4}{5}\right)^6$ | <u>Exponent law used</u><br>1. <i>Negative exponent</i><br>2. <i>Product of powers</i><br>3. <i>Power of powers</i> |

### 4.3.1 WARM-UP: NEGATIVE EXPONENTS

#### TEACHER MOVES

If time permits, have students share with a partner or whole group the expressions they wrote and the strategies they used. Be on the lookout for unique expressions if time is limited for sharing whole group.



#### 4.3.1 Warm-Up: Negative Exponents

- Place a digit in each blue box to make the statement true. Use only digits 0 to 9, at most one time each.

$$\boxed{\phantom{0}} - \boxed{\phantom{0}} = \frac{\boxed{\phantom{0}}}{\boxed{\phantom{0}}\boxed{\phantom{0}}}$$

*Answers will vary.*

$$7^{-2} = \frac{1}{49}$$



4.3.2 Collaboration: Applying More Than One Exponent Law

- Fernanda and Junior were asked to write an equivalent expression of  $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$  by applying multiple exponent laws. Each student first described their plan.
  - Complete each student’s work based on the description of their plan.

| Fernanda                                                                                                                                                                                                                   | Junior                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I will first apply the <b>quotient of powers</b> exponent law. Then, I will apply the <b>power of a power</b> exponent law. Finally, I will re-write the decimal as a fraction and apply the <b>negative</b> exponent law. | I will first apply the <b>power of a quotient</b> exponent law. Then, I will apply the <b>quotient of powers</b> exponent law. Finally, I will re-write the decimal as a fraction and apply the <b>negative</b> exponent law.            |
| $\begin{aligned} &\left(\frac{2.5^5}{2.5^3}\right)^{-1} \\ &= (2.5^2)^{-1} \\ &= (2.5)^{-2} \\ &= \left(\frac{5}{2}\right)^{-2} \\ &= \left(\frac{2}{5}\right)^2 \end{aligned}$                                            | $\begin{aligned} &\left(\frac{2.5^5}{2.5^3}\right)^{-1} \\ &= \frac{(2.5^5)^{-1}}{(2.5^3)^{-1}} \\ &= \frac{(2.5)^{-5}}{(2.5)^{-3}} \\ &= (2.5)^{-2} \\ &= \left(\frac{5}{2}\right)^{-2} \\ &= \left(\frac{2}{5}\right)^2 \end{aligned}$ |

- Explain which method you prefer.  
*Answers will vary. I prefer Fernando’s method because he simplifies the fraction in the second step.*
- Ask your partner which method they prefer. Summarize your partner’s explanation.  
*Answers will vary. I prefer Junior’s method because he distributed the −1 into the fraction, and then applied the quotient rule.*

4.3.2 COLLABORATION: APPLYING MORE THAN ONE EXPONENT LAW

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems for this prompt:

- “The student’s method (or plan) I prefer is...”
- “I prefer this method (or plan) because...”
- “One thing I didn’t like about the other method was...”
- “My partner preferred \_\_\_\_\_’s method. My partner preferred this method because...”

### 4.3.3 ACTIVITY: ROUNDTABLE

#### TEACHER MOVES

Encourage students to discuss the order of steps that are possible for each expression to reinforce that more than one order can be used. Give students time to analyze the prior students' work and discuss if the law was incorrectly applied or if the expression was not equivalent.

#### ENRICHMENT

- What order of steps did you use?
- Why did you choose this order?
- What other order of steps could you have used?
- What strategy do you use to recall all of the exponent laws that could have been used?
- An informal, "get to know you better" question: Which law is your favorite? Explain why.



#### 4.3.3 Activity: Roundtable

In teams of three, assign a number (1–3) to each team member. Each student will complete the **first** step of their assigned problem and identify the exponent law used. Then, pass the problem to the student on your right to check the work and complete the next step in the problem.

For each new step, continue passing the problems to the student on your right until the problems have one base and one exponent. *Answers will vary.*

| Team Member 1:                                                                  |                          |
|---------------------------------------------------------------------------------|--------------------------|
| $\left(\left(\frac{4}{5}\right)^0 \cdot \left(\frac{5}{4}\right)^{-2}\right)^3$ | <u>Exponent law used</u> |
| $\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^2$                   | Negative exponent        |
| $\left(\left(\frac{4}{5}\right)^2\right)^3$                                     | Product of powers        |
| $\left(\frac{4}{5}\right)^6$                                                    | Power of powers          |
| Team Member 2:                                                                  |                          |
| $\frac{(0.5^4)^2}{(0.5^2)^2}$                                                   | <u>Exponent law used</u> |
| $\frac{(0.5)^2}{(0.5)^6}$                                                       | Power of powers          |
| $(0.5)^{-4}$                                                                    | Quotient of powers       |
| $\left(\frac{1}{2}\right)^{-4}$                                                 | Negative exponent        |
| $2^4$                                                                           |                          |
| Team Member 3:                                                                  |                          |
| $\frac{((-3)^2)^4}{(-3)^8 \cdot (-3)^{-2}}$                                     | <u>Exponent law used</u> |
| $\frac{(-3)^8}{(-3)^8 \cdot (-3)^{-2}}$                                         | Power of powers          |
| $\frac{(-3)^8}{(-3)^{-2}}$                                                      | Product of powers        |
| $(-3)^{10}$                                                                     | Quotient of powers       |



4.3.4 Collaboration: Equivalent Expressions

1. Consider the expression  $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$ .

For each expression, work with your partner to determine if it is equivalent to  $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$ . Then, justify your choice by explaining using laws of exponents or showing your work.

| Expression                                                                  | Is it Equivalent to $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$ ? | Justify or Explain                                                                                                              |
|-----------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| a.<br>$\frac{6^{-3}}{6^{-5}} \cdot 6^4$                                     | <input type="radio"/> Yes<br><input type="radio"/> No            | $(\frac{1}{6})^3 = 6^{-3}$                                                                                                      |
| b.<br>$6^{-8} \cdot 6^4$                                                    | <input type="radio"/> Yes<br><input type="radio"/> No            | Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in $6^2$ , which is not equal to $6^{-8}$                |
| c.<br>$(\frac{1}{6})^{-2} \cdot 6^4$                                        | <input type="radio"/> Yes<br><input type="radio"/> No            | Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in $6^2$ , which is equivalent to $(\frac{1}{6})^{-2}$ . |
| d.<br>$(\frac{1}{6})^{-8} \cdot 6^4$                                        | <input type="radio"/> Yes<br><input type="radio"/> No            | Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in $6^2$ , which $\neq (\frac{1}{6})^{-8}$ .             |
| e.<br>$\frac{(\frac{1}{6})^{-3}}{(\frac{1}{6})^5} \cdot (\frac{1}{6})^{-4}$ | <input type="radio"/> Yes<br><input type="radio"/> No            | $6^{-3} \neq (\frac{1}{6})^{-3}$                                                                                                |
| f.<br>$6^6$                                                                 | <input type="radio"/> Yes<br><input type="radio"/> No            | $\frac{6^{-3}}{6^{-5}} \cdot 6^4 = 6^2 \cdot 6^4 = 6^6$                                                                         |

4.3.4 COLLABORATION: EQUIVALENT EXPRESSIONS

TEACHER MOVES

Consider using Sage and Scribe as a strategy. Have a student share their answer as if the expressions are equivalent. Then, have them explain or justify verbally while their partner takes notes on the exponent laws they used to determine if they are equivalent. Then, have partners switch roles. To add complexity, when partners switch roles, the new sage can be asked to explain if the expressions are equivalent using a different order or different series of laws applied.

## 4.3.5 YOUR TURN!

### QUESTIONING

For students who finish early, consider asking them to complete the same problems using a different sequence of steps.



### 4.3.5 Your Turn!

1. Apply the laws of exponents to write an equivalent numerical expression with one base and one positive exponent. Show your steps in the column on the left and identify the law of exponents used in each step in the column on the right. *Answers will vary.*

a.  $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$

| Steps                                                               | Laws of Exponents         |
|---------------------------------------------------------------------|---------------------------|
| $\left(\frac{2}{3}\right)^{15} \cdot \left(\frac{2}{3}\right)^{-6}$ | <i>Power of a product</i> |
| $\left(\frac{2}{3}\right)^9$                                        | <i>Product of powers</i>  |

b.  $\frac{5^4 \cdot 5^4}{5^3}$

| Steps             | Laws of Exponents         |
|-------------------|---------------------------|
| $\frac{5^8}{5^3}$ | <i>Product of powers</i>  |
| $5^5$             | <i>Quotient of powers</i> |



#### 4.3.6 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of today's learning target.

- ☐ I can apply the Laws of Exponents to generate equivalent numerical expressions.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

*Answers will vary.*

### 4.3.6 WRAP-UP: REFLECTION

#### TEACHER MOVES

Prompt students to identify problems from the lesson that illustrate evidence of their level of understanding. If they are not confident with the learning target, ask them to identify which question(s) they struggled with. This information could be used to pair students in the next lesson. Students who are confident could help their peers who are struggling with particular problems.

#### DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.